23.10.2026

Method for GW190425 Optical Counterpart Search in ZTF Survey Data

Efremova M. A., Pozanenko A. S., Pankov N. S., Volnova A. A., Moskvitin A. S.



The XXVIII DAMDID conference

Gravitational Wave Observatories

LIGO Hanford
Washington, USA



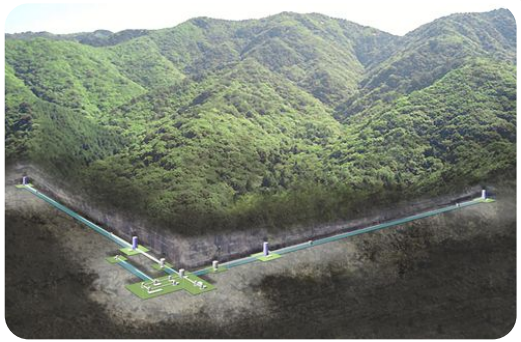
LIGO Livingston
Louisiana, USA



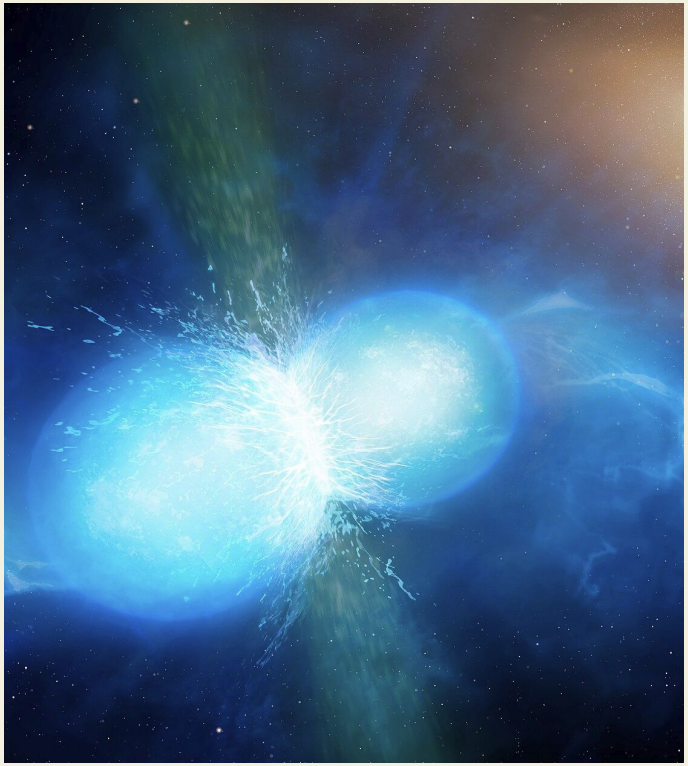
VIRGO
Cascina, Italy



KAGRA
Kamioka, Japan



The Two Binary Neutron Star Mergers



GW190425

DISTANCE

159 (+69 / −71) Mpc

TOTAL MASS

3.4 (+0.3 / −0.1) $M_\odot$

[B. P. Abbott et al., 2020]

APPARENT MAGNITUDE

NOT DETECTED

April 2019

GW170817

DISTANCE

43.8 (+2.9 / −6.9) Mpc

TOTAL MASS

2.74 (+0.04 / −0.01) $M_\odot$

[B. P. Abbott et al., 2017]

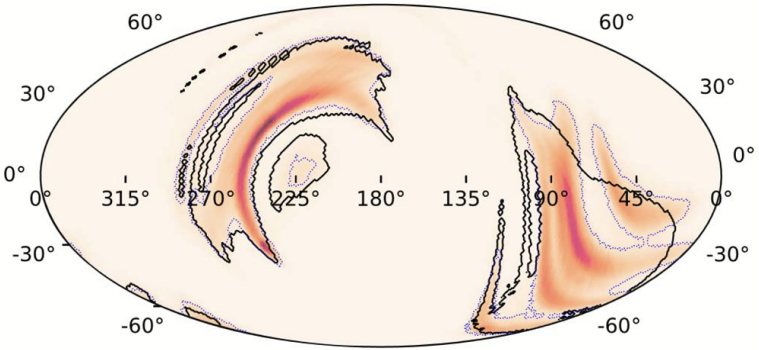
APPARENT MAGNITUDE

17.46 ± 0.03 (r-band)
[S. Valenti et al., 2017]

August 2017

...And Their Localization Areas

GW190425

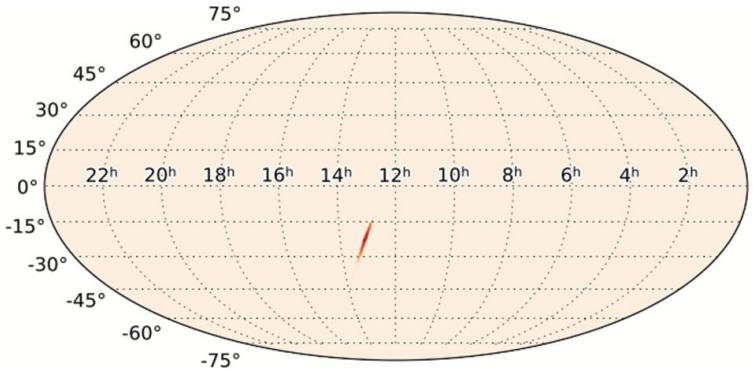


LOCALIZATION AREA

8,284 deg²

[B. P. Abbott et al., 2020]

GW170817



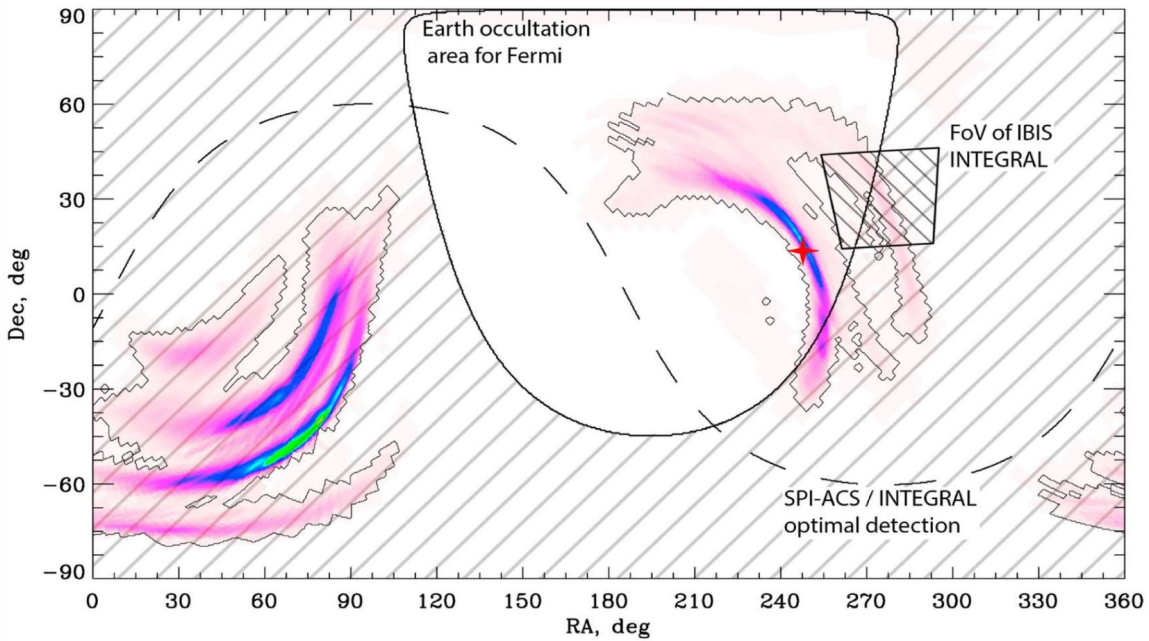
LOCALIZATION AREA

28 deg²

[B. P. Abbott et al., 2017]

GW190425 Gamma Ray and Radio Counterparts?

GW + Gamma-Ray + Radio Localization Sky Map



[A. E. Rodin et al., 2026]

INTEGRAL (Gamma)



BSA (Radio)



The Optical Telescope that Observed the Area: ZTF

Palomar Observatory



© Palomar/Caltech

ZTF Specifications & Alerts

ZTF FACTS

- Cadence of ~2 days
- 5σ detection limit of 20.7 mag
- Raw data is processed in real time
- Alerts are freely available immediately

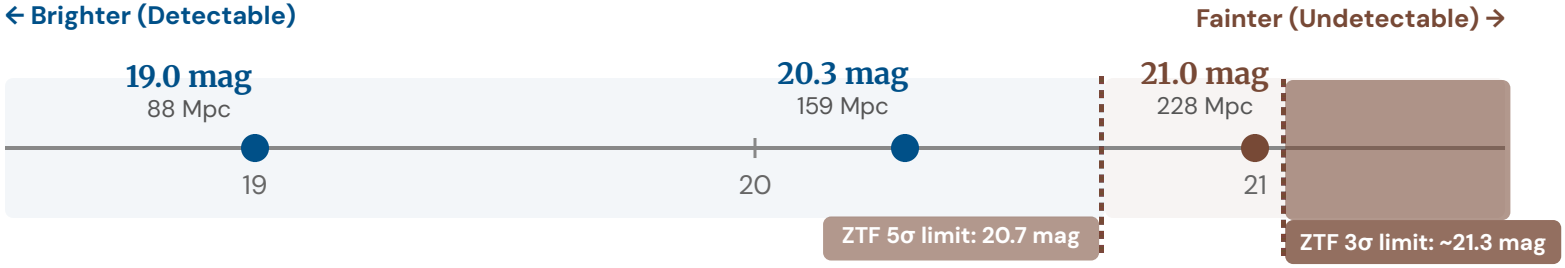
TRANSIENT ALERT DEFINITION

A ZTF transient alert is defined as a **5σ change** in brightness in the image relative to the reference epoch.

What Should ZTF Have Seen?

NEAR EDGE OF DISTANCE RANGE	BEST–ESTIMATE DISTANCE	FAR EDGE OF DISTANCE RANGE
19.0 mag	**20.3 mag**	**21.0 mag**
at 88 Mpc	at 159 Mpc	at 228 Mpc

Expected r-band brightness of a GW170817-like kilonova



GW170817 (17.46 mag at 43.8 Mpc) scaled by distance only, $m = 17.46 + 5 \log_{10}(d / 43.8\ \text{Mpc})$
no extinction, K-correction or time evolution.

Previous Work using ZTF Data

REFERENCE STUDY (GROWTH / ZTF)

GROWTH on S190425z: Searching thousands of square degrees to identify an optical or infrared counterpart to a binary neutron star merger with the Zwicky Transient Facility and Palomar Gattini IR

Michael W. Coughlin,[1] Tomás Ahumada,[2] Shreya Anand,[1] Kishalay De,[1] Matthew J. Hankins,[1]

ALERT VOLUME

338,646

ZTF TRANSIENT ALERTS

SEARCH OUTCOME

0

VIABLE COUNTERPARTS

Our Method: How is It Different?

01. DIFFERENCE DATA

Check **all difference data** and not only alerts

02. GALAXY CATALOG

Create a **combined galaxy catalog**

03. GALAXY CUTOUTS

Use targeted **galaxy cutouts**

04. DETECTION THRESHOLD

$5\sigma \rightarrow 3\sigma$

Lower detection threshold to capture faint signals

05. MACHINE LEARNING FILTER

Apply **ML algorithm "BRAAI"** to score real/bogus for objects on difference images

[Duev+ 2019]

Combined Galaxy Catalog

COMBINED DATA SOURCES

- SDSS-DR16
- DESI-DR8
- GLADE+
- NED-LVS
- CatWISE
- GAIA-DR3

FILTERING CRITERIA

FILTERED TO INCLUDE ONLY OBJECTS:

- In the localization area
- Similar to galaxies
- With a distance either in the range of GW190425, or unknown

26,849

TOTAL NUMBER OF GALAXIES

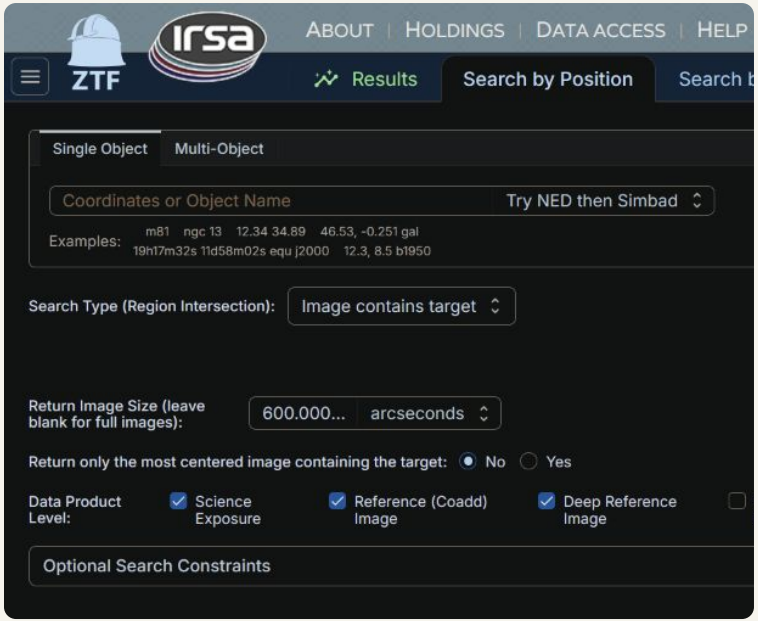
14,264

GALAXIES WITH KNOWN DISTANCES

ZTF Images

DATA ACCESS & INTERFACE

Data can be freely accessed from the web interface or API Queries

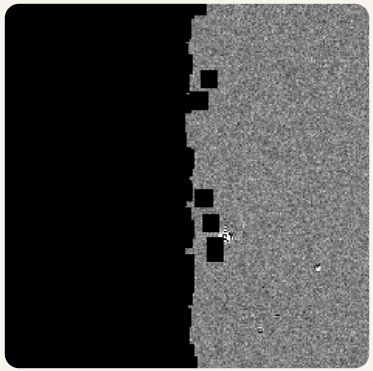


DIFFERENCE IMAGES & QUALITY

~58k Difference Cutouts

25–26 of April (first 2 days after trigger)

Issue: Masked Images



The Candidate Seeking Pipeline

START: 26,849 galaxies

STEP 01

Signal to Noise Ratio

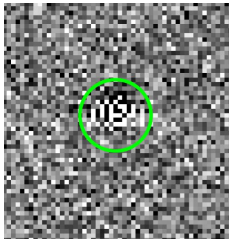
631 galaxies

$$SNR = \frac{\mu_{signal}}{\sigma_{noise}} \geq 3$$

STEP 02

Artefact Filtering

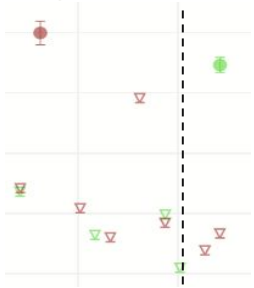
215 galaxies



STEP 03

No pre-trigger detection

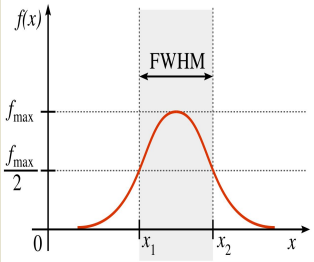
14 galaxies



STEP 04

FWHM Consistency

4 galaxies



STEP 05

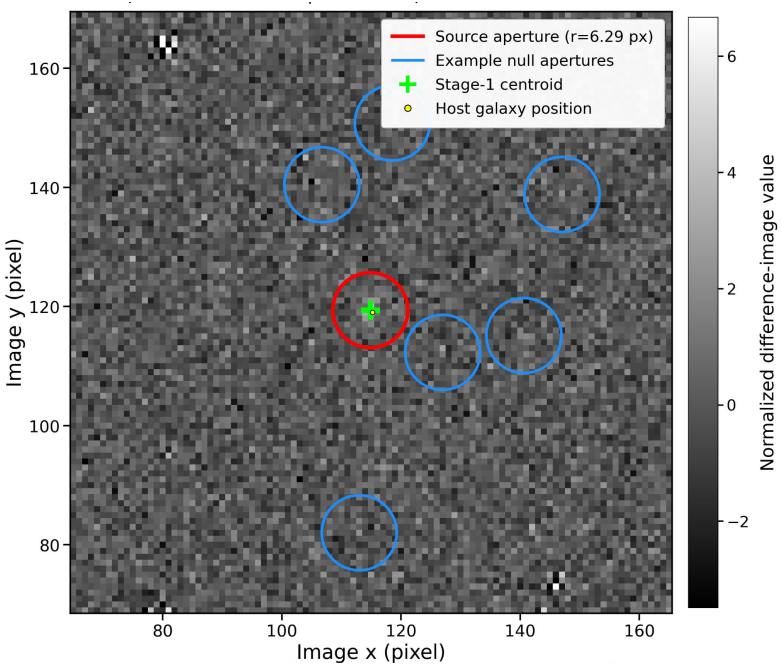
Asteroid Check

3 galaxies



Step 1: Signal / Noise Ratio

FITS APERTURE PHOTOMETRY



$$SNR = \frac{F_{source} - \overline{F_{null}}}{\sigma_{null}}$$

SNR MEASUREMENT ALGORITHM

01 | Centroid Localization

Aperture Radius r = 2 × FWHM

02 | Flux & Background

Measure aperture flux & background μ_{bg}, σ_{bg}

03 | Empirical SNR Calculation

Calculate {Fnull} in 100 random apertures

04 | Decision Thresholds

SNR ≥ 3 ⇒ **Detection**, else **Upper Limit**

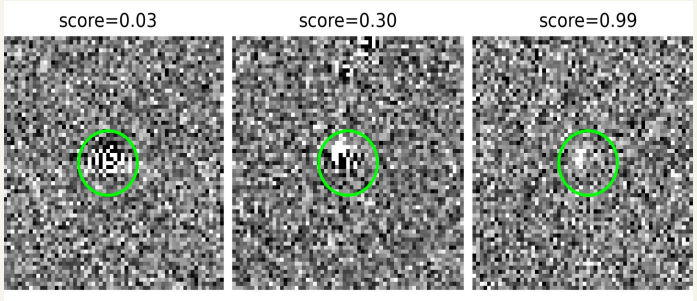
Step 2: Artefact Filtering (BRAAI)

Step 2 of 5 · 631 → 215

METHODOLOGY & EXAMPLES

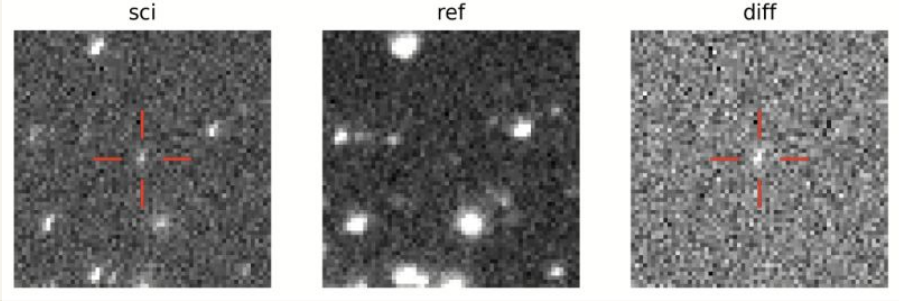
- **braai CNN**: Deep learning model designed for ZTF subtraction artefacts.
- **Realness Score**: Output range [0.0, 1.0]; higher score = higher probability of real transient.
- **Threshold Selection**: Cut value set at 0.25.

BRAAI Score Examples

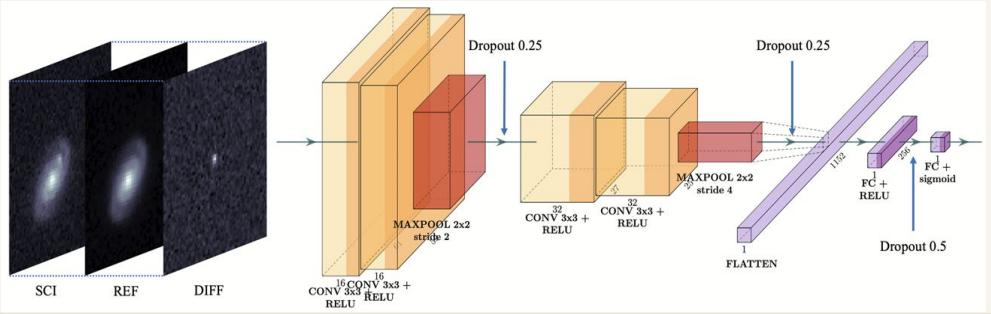


REAL/BOGUS ML ARCHITECTURE

Input Format: 63×63 px cutout triplet (Science, Template, Difference)



Real/Bogus ML Architecture

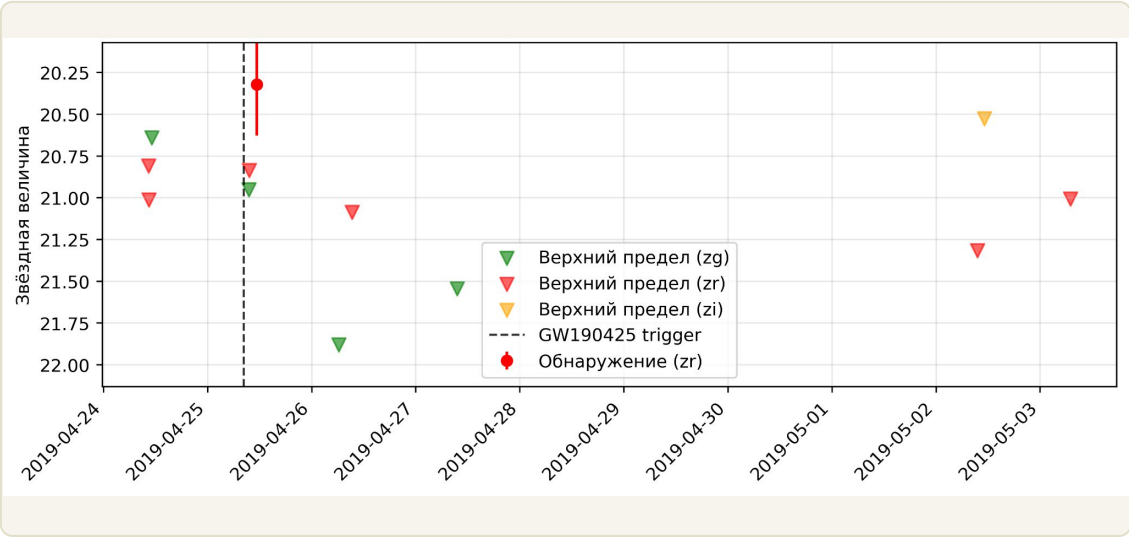


Step 3: Pre-Trigger Images

SELECTION CRITERIA & METHODOLOGY

- **Time Periods:** SNR for each object is measured 10 days pre- and post trigger.
- **Transient Timing:** Must appear no earlier than the GW trigger event.
- **Rejection Rule:** Significant S/N prior to April 25 marks candidate as unassociated with GW190425.

CANDIDATE LIGHT CURVE



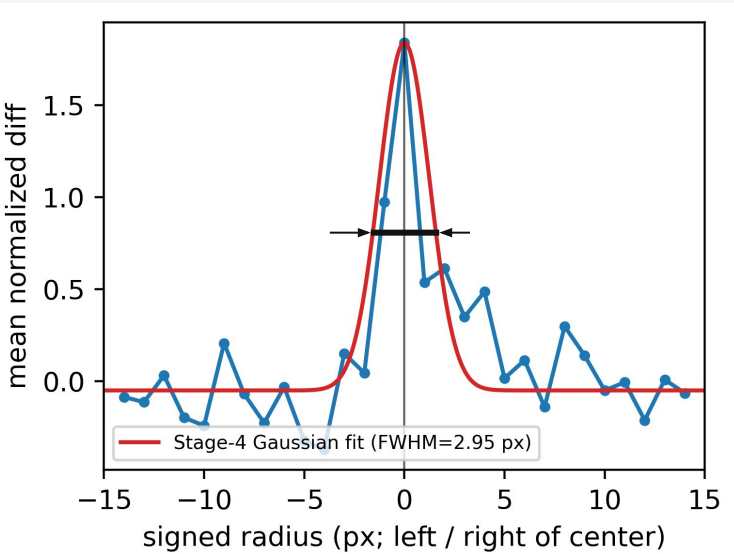
Step 4: FWHM Consistency

METHODOLOGY & FWHM CRITERIA

FWHM Filtering

- **2D Gaussian Profile:** A two-dimensional Gaussian profile is fitted around the source position to calculate FWHM.
- **Header Comparison:** The measured FWHM is compared against the value in the image header.
- **Rejection Threshold:** If the calculated FWHM deviates by more than **50%** from the header value, the candidate is rejected.

GAUSSIAN PROFILE FIT



Step 5: Asteroid Filtering

METHODOLOGY & SKYBOT CHECK

- **SkyBoT Query:** For each candidate, a query is submitted to the IMCCE SkyBoT service.
- **Search Cone:** The search is conducted within a cone with a radius of **5 arcminutes**, considering ZTF observatory coordinates.
- **Filtering:** If a known asteroid or comet is found in the area, the candidate is flagged as a Solar System object.

IDENTIFICATION & PARAMETERS

5′
Search Radius Cone

ZTF
Observatory Coords

IMCCE SkyBoT

Sky Body Tracker service for identifying known Solar System objects.

Manual checks:

IAU The International Astronomical Union Minor Planet Center

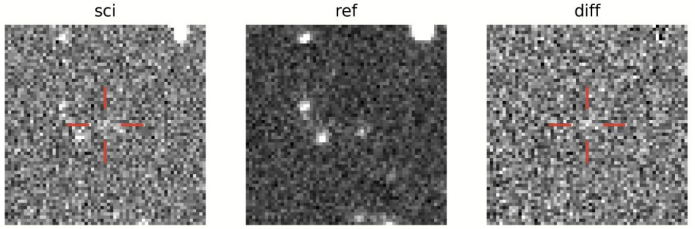
Results

SUMMARY & CANDIDATE SELECTION

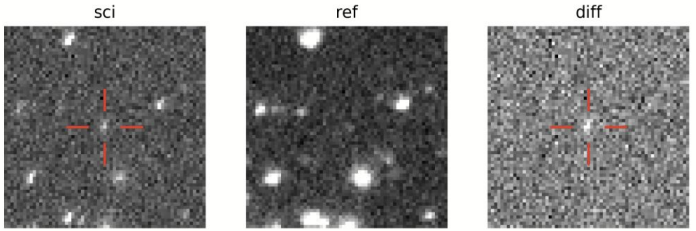
Key Results

- **Step 4:** Selected **4 objects** from 58,289 difference images.
- **Signal Parameters:** All objects have low SNR (3.1–4.5) and braai scores of 0.9–1.0.
- **Single Filter:** All candidates have been detected in one filter only (ZTF-r).
- **Step 5:** Object #3 is **asteroid 2004 OH6** (filtered out).
- **Current Status:** The remaining 3 objects are classified as low-significance candidates.

CANDIDATE TRIPLETS (SCI / REF / DIFF)



Candidate #2: Transient Object (r-filter)



Object #3: Asteroid (2004 OH6)

Conclusions & Plans

CONCLUSIONS & KEY FINDINGS

01 ZTF Search Pipeline
Developed search method for GW190425 optical counterpart using ZTF data (SNR ≥ 3) across **26,849 galaxies**.

02 Candidate Selection
Filtered candidate selection yielded **3 candidates** from difference images.

03 Cross-Identification
None confidently identified as GW190425 counterpart.

FUTURE PLANS & NEXT STEPS

01 No Galaxy Catalog
Examine the full radio localization area without sticking to galaxy cutouts only.

02 Full Localization Region
Apply the developed search methodology across the entire GW190425 localization area.

03 Multi-Event Application
Extend the processing pipeline to other gravitational-wave events in ZTF survey data.

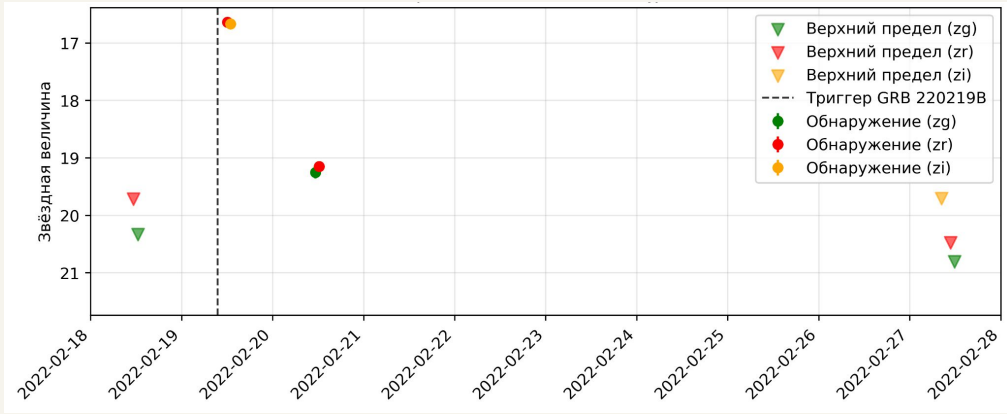
(BACKUP) Trial Run: ZTF22aabjpxh

METHODOLOGY & TRIAL RESULTS

Verification Target

- **Target Object:** Optical transient ZTF22aabjpxh linked to GRB 220219B.
- **Redshift & Search:** Photometric z = 0.293; searched 1.25 deg² region around host galaxy.
- **Top Rank:** Achieved highest SNR and realness (braai) score among all 14 candidates.
- **Filter Metrics:** r-filter SNR = 99.98 (braai = 1.0); i-filter SNR = 96.18 (braai = 0.99).

LIGHT CURVE



(BACKUP) ZTF Alert Search in Radio Area

RESULTS

2 OBJECTS FOUND

- **ORPHAN:** checked through MPC; identified as a Near-Earth object (2019 JS5)
- **VS:** Variable Star; multiple detections pre-trigger (January, March 2019)

LIGHT CURVE

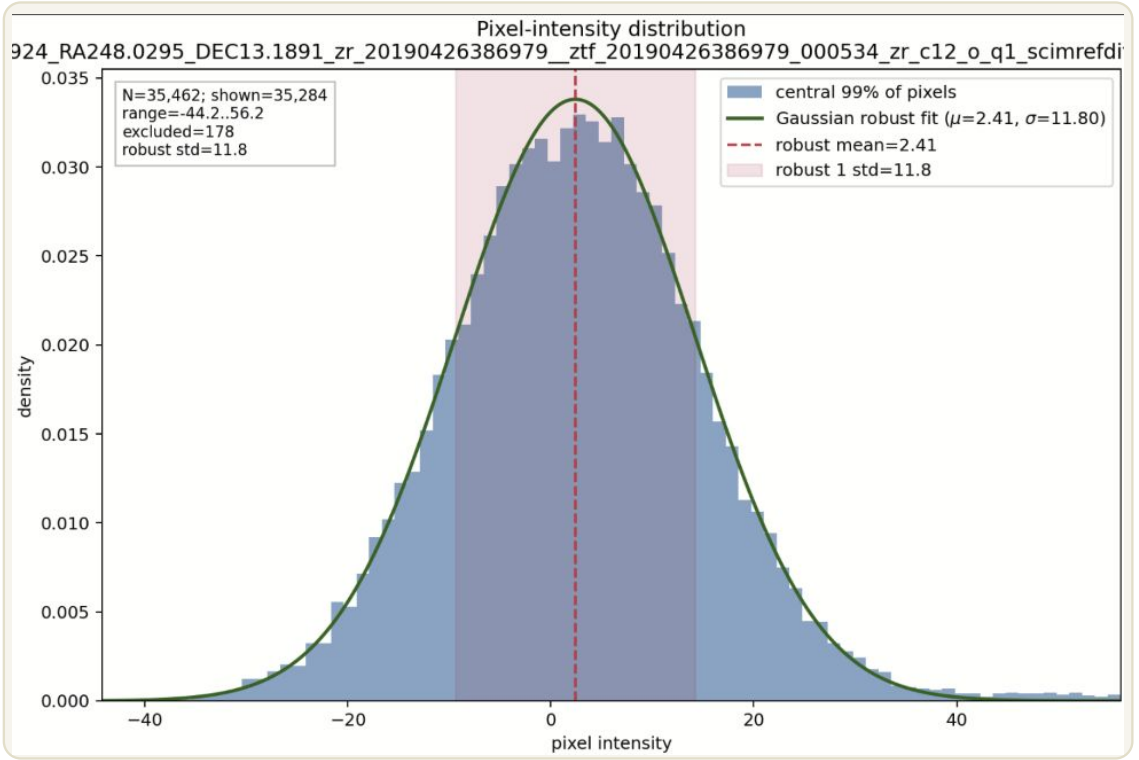


(BACKUP) Non-Poisson Distribution

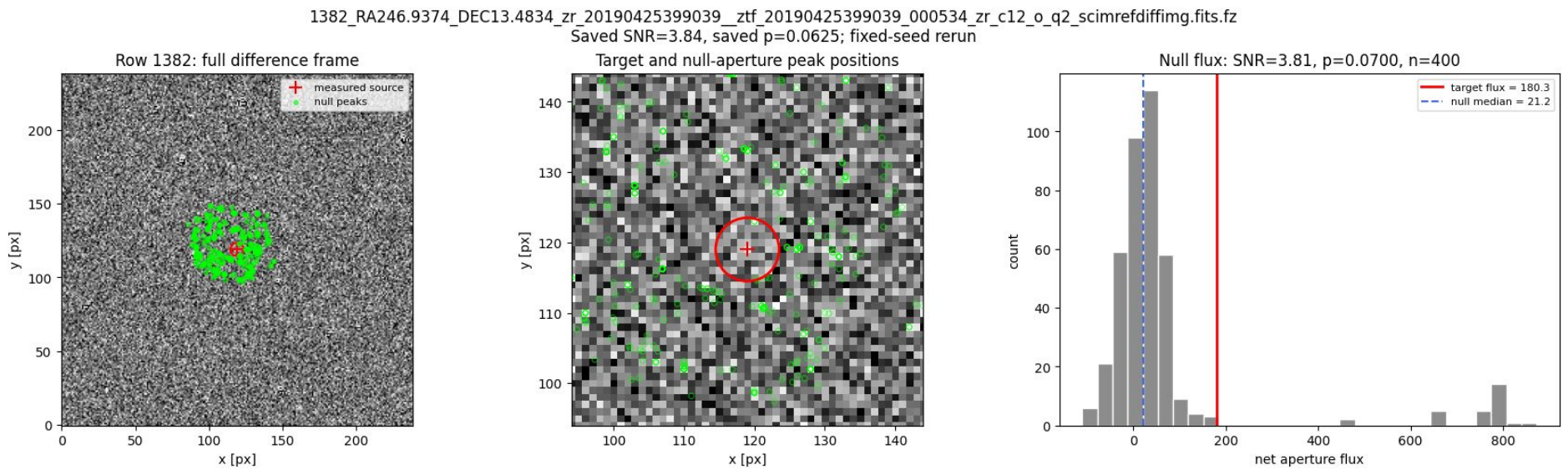
RESULTS

The non-poissonian statistics of difference images is the motivation for the usage of empirical SNR.

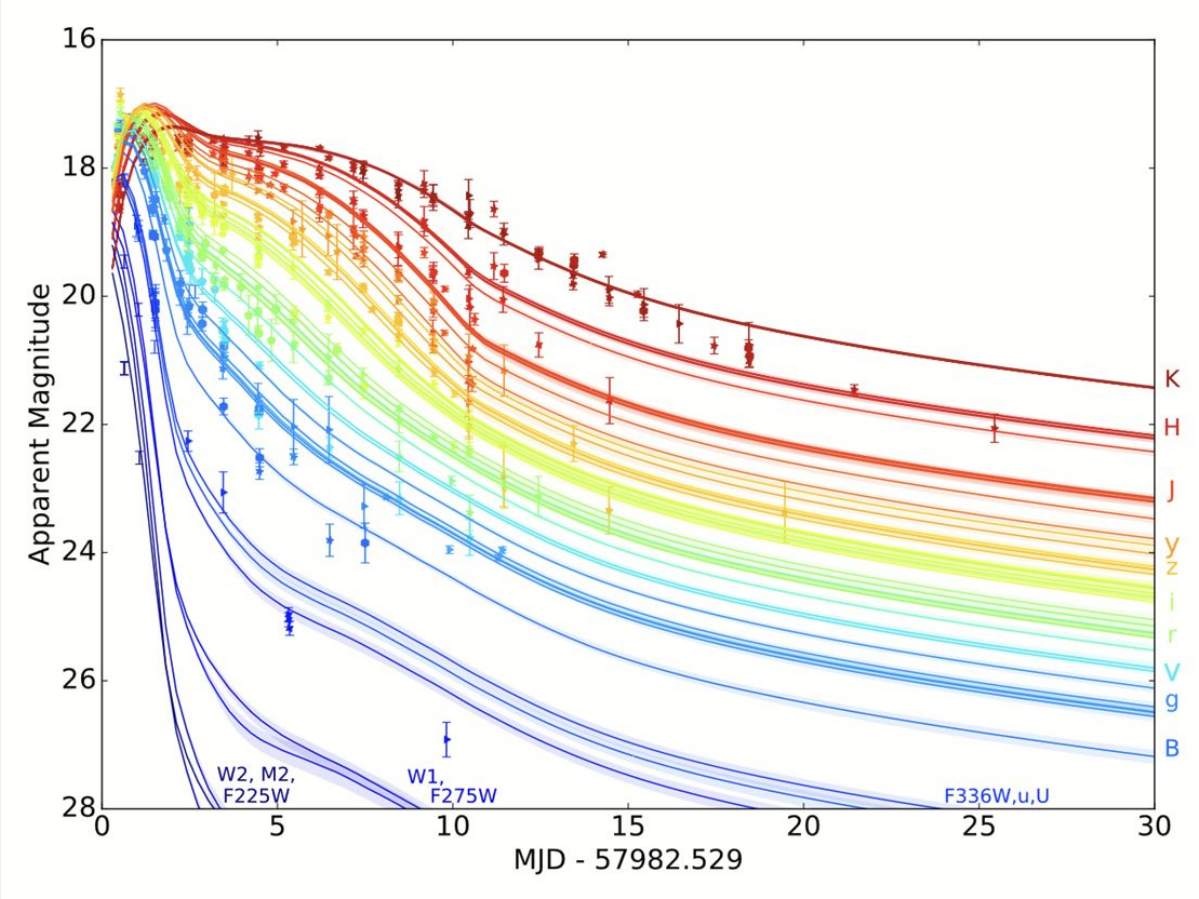
DIFFERENCE IMAGE HISTOGRAM



(BACKUP) Non-Poisson Distribution & SNR



(BACKUP) GW170817 Light Curve



[Villar et al., 2017]

(BACKUP) Step 1: Signal / Noise Ratio

METHODOLOGY & PARAMETERS

FITS & Aperture

- **FITS Format**: Stores pixel brightness values.
- **FWHM**: Characterizes source size; sets circular aperture radius $r = 2 \times \text{FWHM}$.

Centroid & Background

- **Quadratic Centroid**: Subpixel refinement of source coordinates (cx, cy).
- **Sigma-Clipping**: Estimates background μbg and noise σbg, robust to bright stars.

Input & Output Data

- **Input**: Difference FITS image, galaxy center.
- **Output**: Centroid position, flux F, error, standard & empirical SNR.

SNR MEASUREMENT ALGORITHM

01 | Initialization & Centroid

Extract FWHM; set radius $r \leftarrow 2 \times \text{FWHM}$. Locate centroid ($cx$, cy) within $4 \times$ FWHM region.

02 | Measurement & Sigma-Clipping

Calculate flux F in aperture r. Exclude central source to estimate background μbg and noise σbg.

03 | Standard & Empirical SNR

Compute net flux $F\text{net} = F - \mu\text{bg} \cdot N\text{pix}$. Estimate SNRemp using 100 random apertures (4–12 FWHM).

04 | Decision Rules & Thresholds

- If $2 \leq \text{SNRemp} \leq 6$: refine calculation using 400 apertures.
- If $\text{SNRemp} \geq 3$: **Detection**, otherwise **Upper Limit**.

(BACKUP) Candidate Table

Величина	Кандидат 1	Кандидат 2	Кандидат 3	Кандидат 4
RA (deg)	247.959481	247.237205	247.113962	247.223780
Dec (deg)	13.081366	13.273388	13.152279	13.089430
SNR	4.5	3.8	3.8	3.1
braai	1.0	1.0	1.0	0.9
Source m	20.3 ± 0.3	20.6 ± 0.3	20.2 ± 0.3	20.7 ± 0.3
SDSS r	—	—	22.3	21.0
$W1$	—	—	18.5	—
$W2$	—	—	18.2	—
DESI g	24.0	23.9	22.8	21.6
DESI r	23.4	23.2	21.9	21.0
DESI z	23.6	23.1	21.4	20.7
DESI offset (")	0.01	0.00	0.38	0.31
D_L (Mpc)	3666	3630	2998	1147
D_L lower (Mpc)	< 1	< 1	206	< 1
D_L upper (Mpc)	11316	11325	6597	3292
FWHM (px)	3.37	2.95	2.54	2.42

Блеск источника приведён по разностному изображению ZTF в r.
SDSS r — блеск в фильтре r по SDSS.
$W1$, $W2$ — блеск в полосах W1 и W2 по CatWISE.
DESI g, r, z и DESI offset — фотометрия и смещение по данным DESI-DR9; offset — в угловых секундах.
D_L — расстояние до галактики; D_L lower/upper — нижние и верхние пределы на расстояние с учетом ошибки, Мпк.
FWHM — полная ширина на половине максимума, пиксели.
Прочерк (—) обозначает отсутствие данных.